MCQ Quick-Ref: ggplot2 vs. Plotly

Essential Technical Concepts for Exams & Interviews

1. Architecture & Philosophy

- **ggplot2 (R/plotnine):** Based on *Leland Wilkinson's* "Grammar of Graphics." It treats plots as a series of stacked layers.
- **Plotly (Cross-platform):** Built on *d3.js* and *stack.gl*. It is web-native and JSON-based.
- **Stat-vs-Interactive:** ggplot2 is primarily for **declarative static** graphics. Plotly is for **imperative interactive** graphics.

2. Key Syntax Components

ggplot2 Layering Order:

1. **Data:** The data frame.
2. **Aesthetics (aes):** Mapping variables to axes (x, y), color, size, and shape.
3. **Geoms:** Geometric objects (`geom_point`, `geom_line`).
4. **Facets:** Sub-plots (`facet_wrap`, `facet_grid`).

Plotly Express (Python) Categories:

- **px.scatter / px.line:** High-level wrapper functions.
- **Graph Objects (go):** Low-level control (`go.Figure`, `go.Scatter`).
- **Layout:** Global styling (titles, margins, hovermodes).

3. Critical "MCQ Traps"

- **The Plus Sign:** ggplot2 uses + to add layers. Plotly (in R) often uses the pipe %>% and in Python uses methods like `.update_layout()`.
- **Global vs Local aes:** In ggplot2, an aesthetic defined in `ggplot()` is inherited by all geoms; an aesthetic defined inside a `geom_*()` is local to that layer only.
- **The "Magic" Link:** `ggplotly(p)` is the R function that converts a static ggplot2 object into an interactive Plotly object.
- **Rendering:** ggplot2 outputs to "Devices" (PNG, PDF). Plotly outputs to "Renderers" (HTML, Browser).

4. Comparison Table

Feature	ggplot2	Plotly
Philosophy	Grammar of Graphics	Web-ready / D3.js
Interactivity	Static only (mostly)	Interactive by default
3D Plots	Not native (extension)	Native (`scatter3d`)
Language	R (primary)	Python, R, JS, Julia
Zoom/Pan	Export at resolution	Built-in UI controls
Custom Themes	`theme_bw()`, etc.	`template='plotly_dark'`

5. Common Mapping (Naming)

ggplot2 Term	Plotly Equivalent
`geom_point()`	`px.scatter()` / `go.Scatter(mode='markers')`
`geom_bar()`	`px.bar()`
`geom_histogram()`	`px.histogram()`
`geom_boxplot()`	`px.box()`
`facet_wrap()`	`facet_col` / `facet_row` arguments
`labs(title=...)`	`update_layout(title=...)`

6. Scalability

- **Large Datasets:** ggplot2 handles millions of points better in static PDF/PNG. Plotly can slow down browsers if too many SVG elements are rendered (use `scattergl` for $> 100k$ points).
- **Scientific Publishing:** ggplot2 is the industry standard for academic journals due to precise typography and vector output.